

Code-Layout, Readability and Reusability

Date	15 JULY 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Uniform formatting throughout the project
2	Proper File Structure	Files and folders are logically organized	Yes	Organized frontend, backend, and database folders
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables clearly describe their purpose
4	Function / Method Names	Functions are descriptively named	Yes	Functions use descriptive naming conventions
5	Code Comments	Inline and block comments explain logic	Yes	Important logic is documented
6	Modular Design	Code is split into reusable functions/modules	Yes	Independent modules for authentication, loans, AI, and database
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code minimized through reusable functions
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Exceptions handled for login, database, and AI operations

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Authentication Module	Flask, JWT	Login & Registration	High
2	Loan Management Module	Python, Flask	Loan Operations	High
3	Financial Profile Module	Python	Financial Analysis	High
4	AI Prediction Module	Scikit-learn	Settlement Prediction	High
5	Database Module	PostgreSQL	All Backend Services	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project architecture
Readability	5	Easy-to-understand naming conventions
Reusability	5	Modular and reusable components
Documentation / Comments	5	Adequate comments for maintainability
Overall Score	5/5	Production-ready and scalable codebase